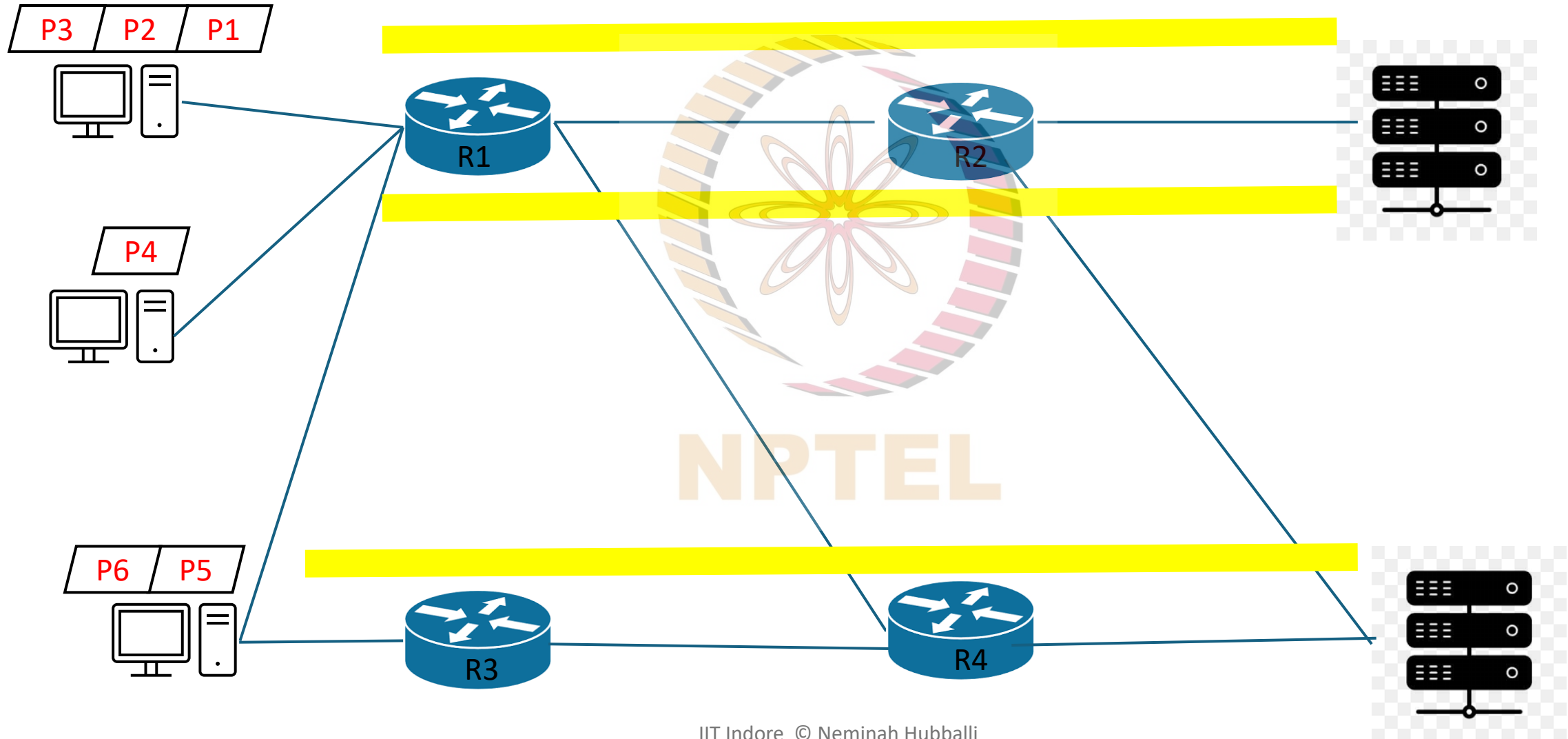
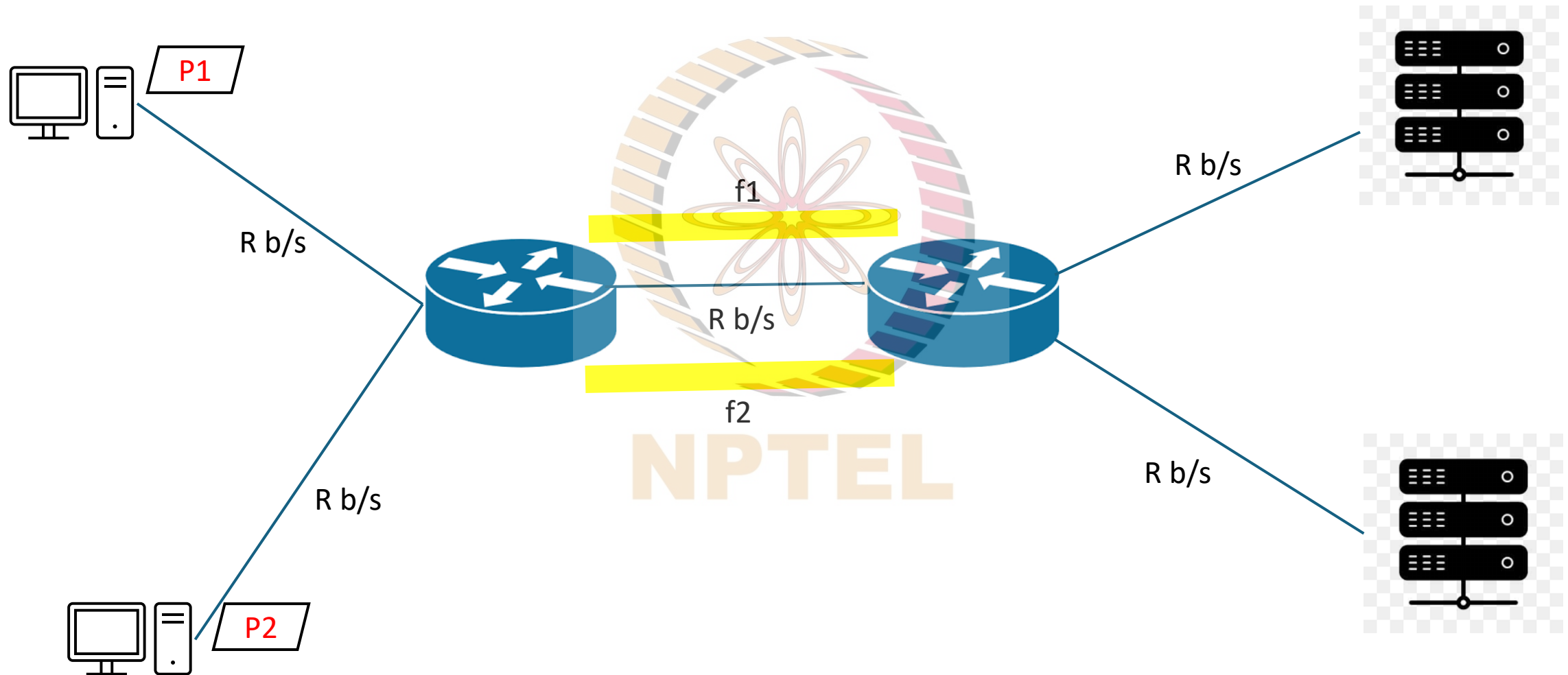


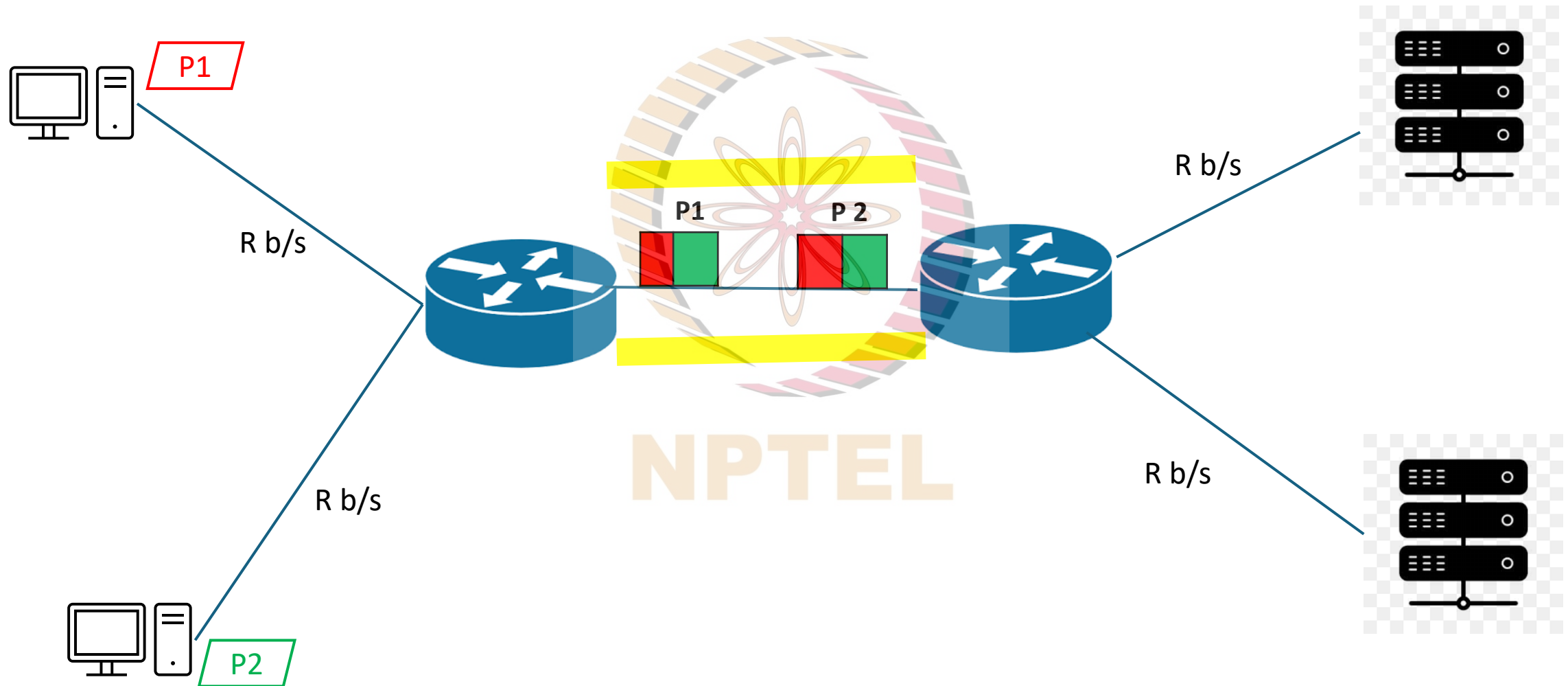
# Circuit Switching



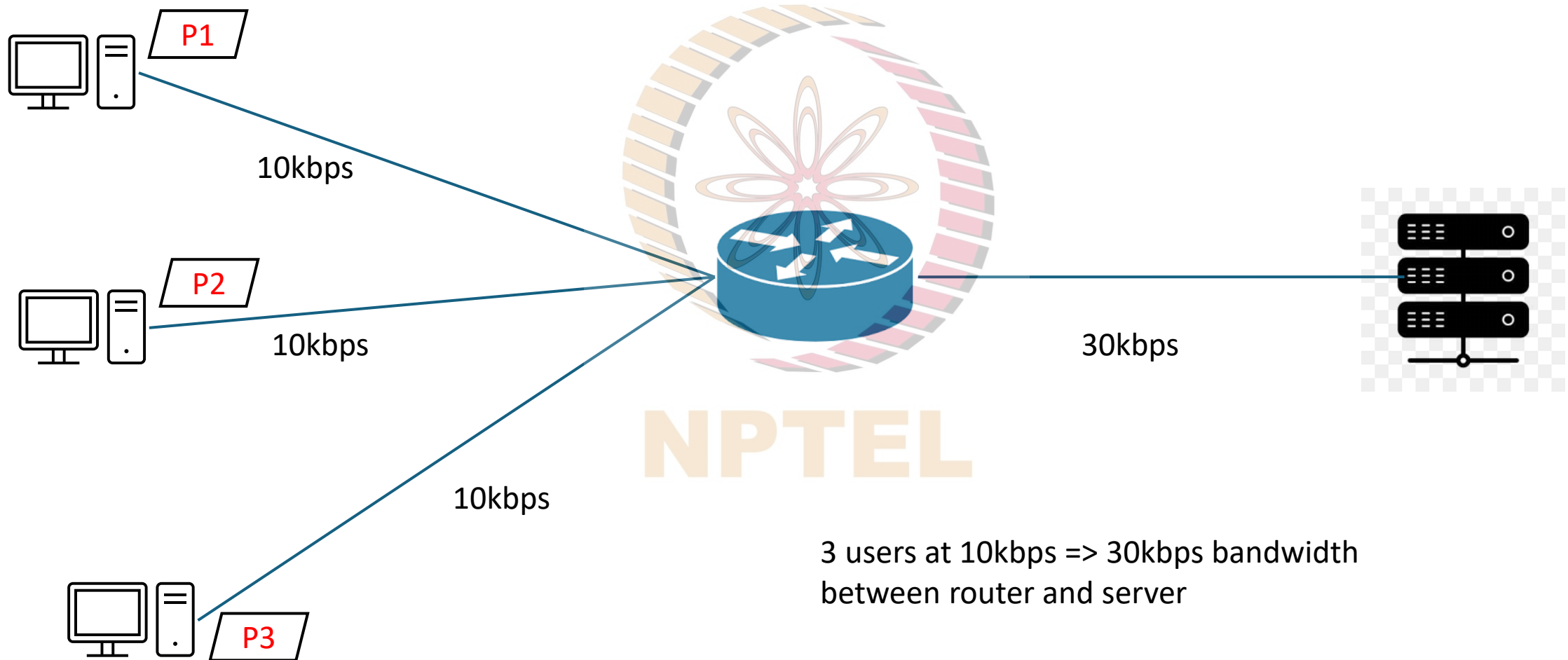
# Multiplexing in Circuit Switching with FDM



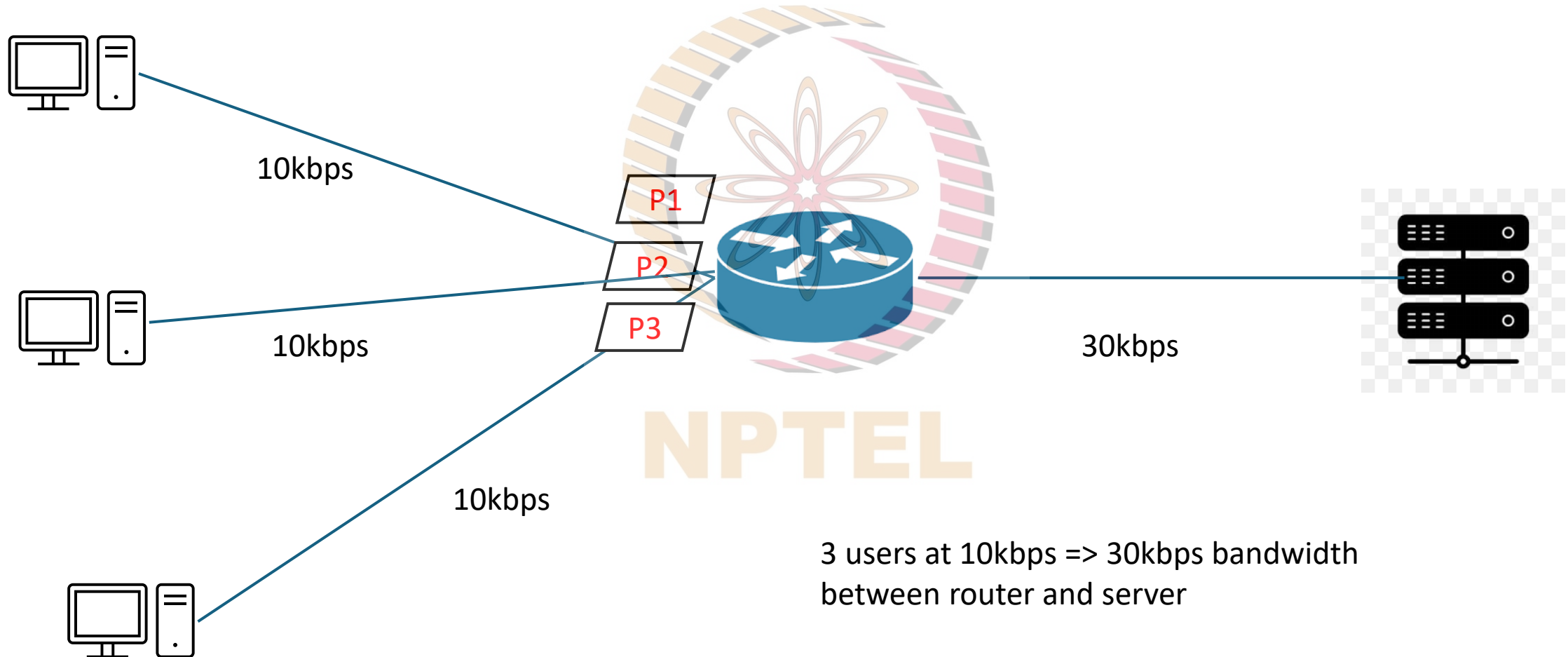
# Multiplexing in Circuit Switching with TDM



# How to Support More Users ?



# How to Support More Users ?



# Transmission Over Circuit Switched Networks

Q 1: A file of 6,40,000 bits has to be transmitted from Host A to Host B over a circuit-switched network. Suppose that all links in the network use TDM with 24 slots and have a bit rate of 1.536 Mbps. Also suppose that it takes 500 msec to establish an end-to-end circuit before Host A can begin to transmit the file. How long does it take to send the file?

Soln: Each circuit transmission rate  $1.536 \text{ Mbps}/24 = 1.536 \times 10^6/24 = R$   
bps = 64,000 bps (or 64 kbps)

File has 6,40,000 bits  $\rightarrow L = 6,40,000$  bits

$$T_{total} = \text{Setup Time} + d_{trans}$$

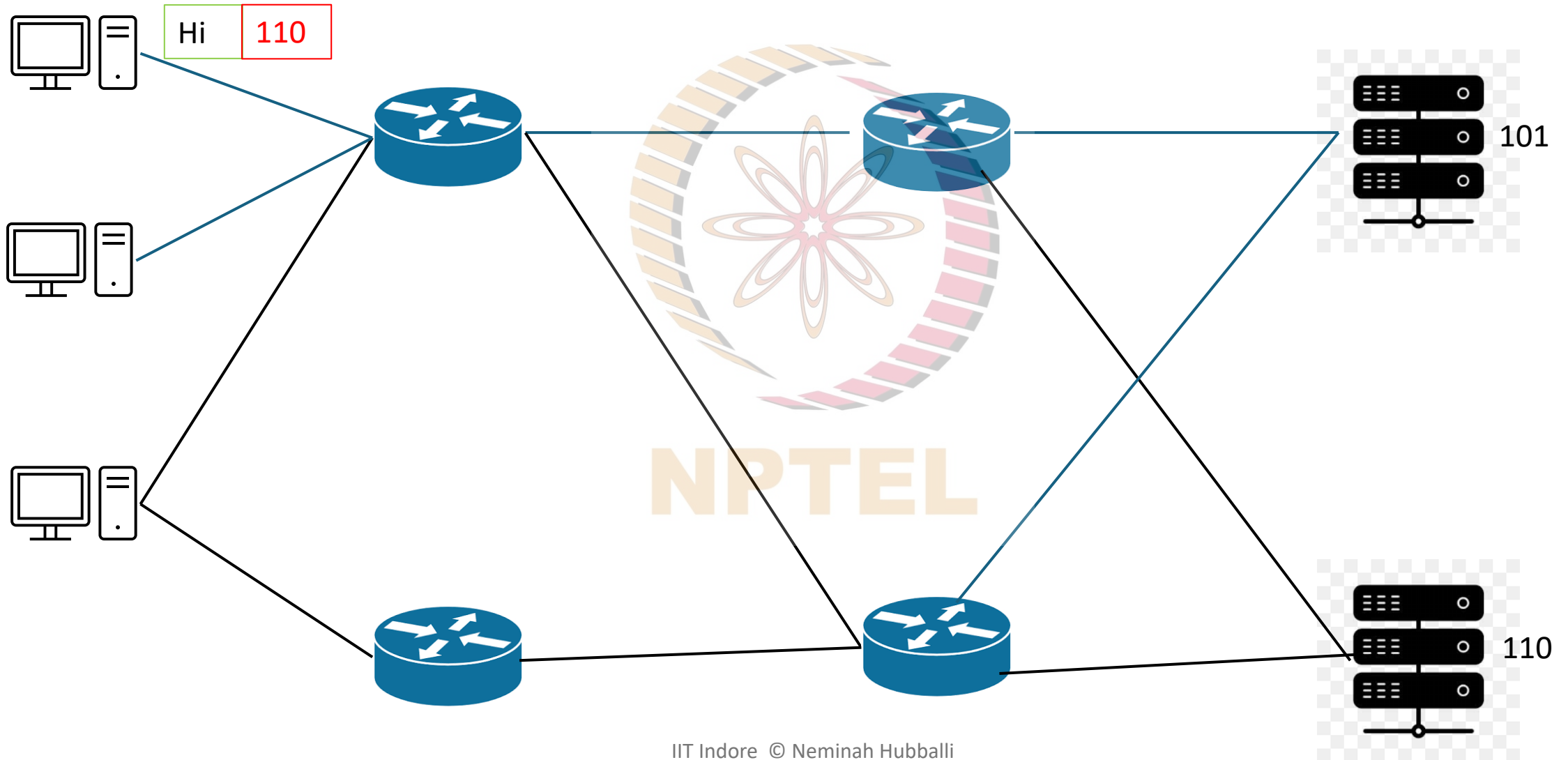
$$\text{Transmission time} = \frac{L}{R} = 10 \text{ sec}$$

$$T_{total} = 0.5 \text{ s} + 10 \text{ s} = 10.5 \text{ seconds}$$

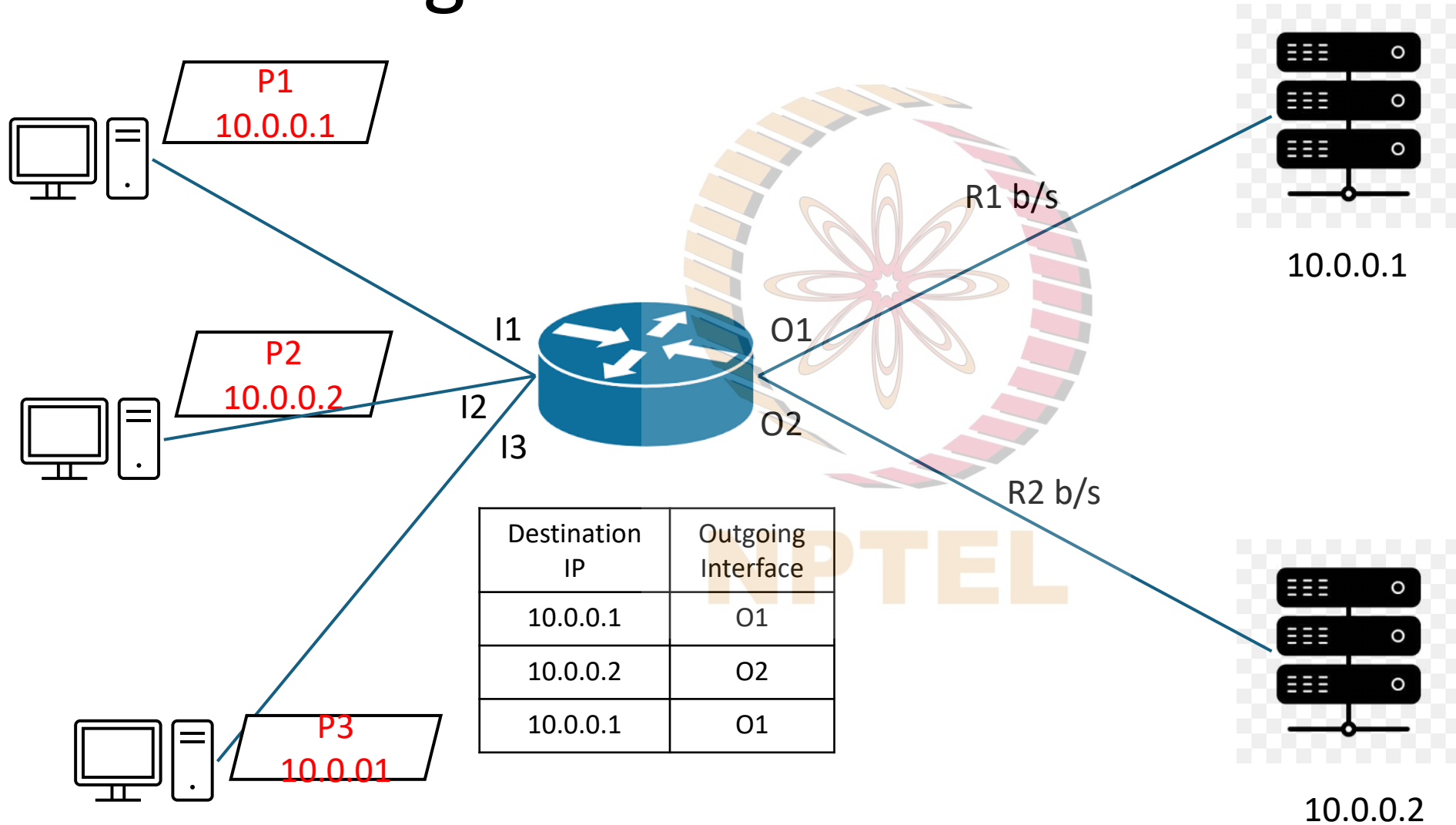
# Can we do Better ?

- ☐ Suppose a user transmits only 20% of the time
  - ☐ Leads to 80% wastage of bandwidth
- ☐ Pertinent questions
  - ☐ If other 2 users require more bandwidth, can we use this 80% ?
  - ☐ If all 3 users are lazy and are active only 20% of the time, can we support more users ?
  - ☐ Can we avoid the burden of connection setup?

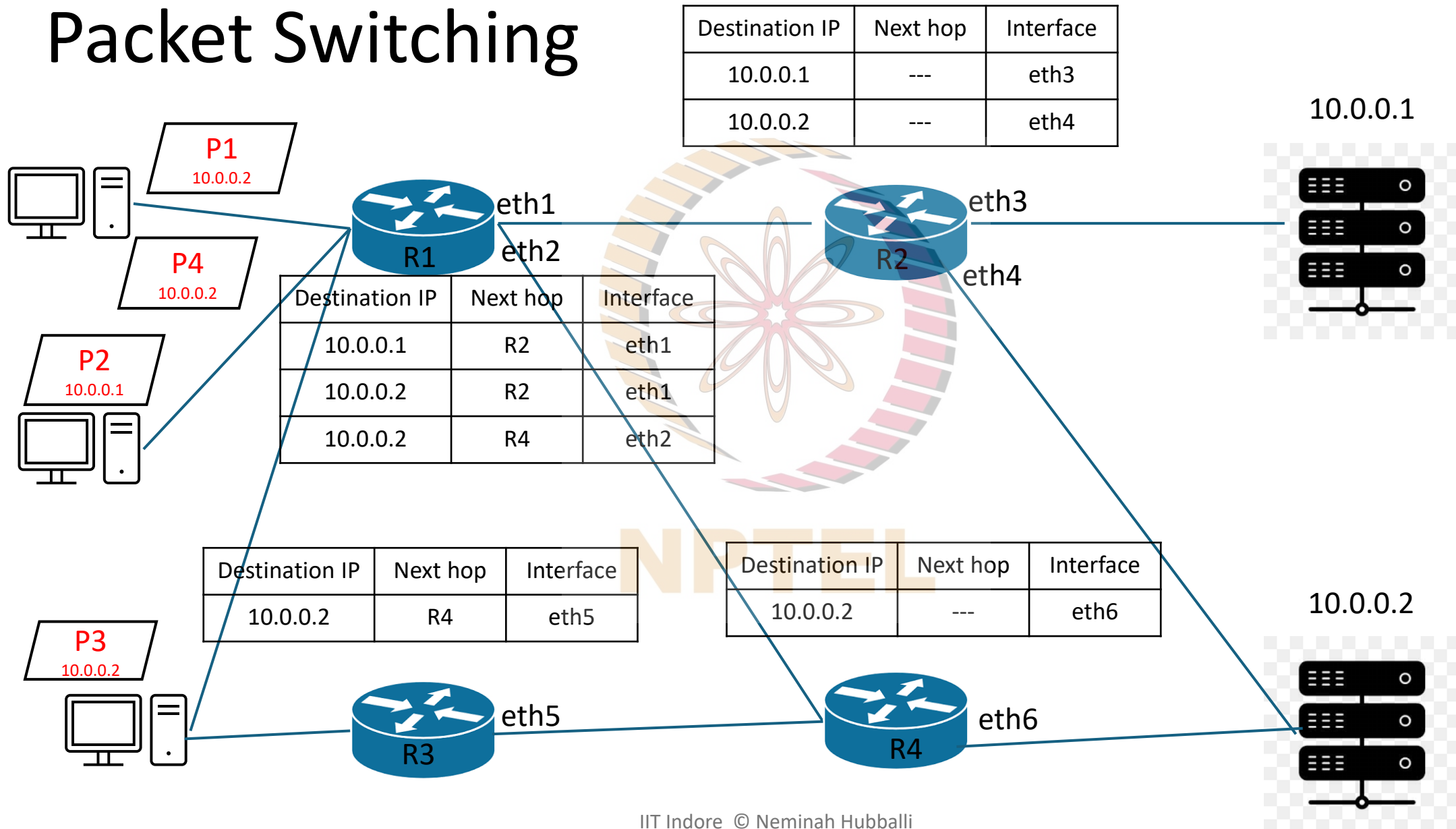
# Packet Switching Concept



# Forwarding Tables

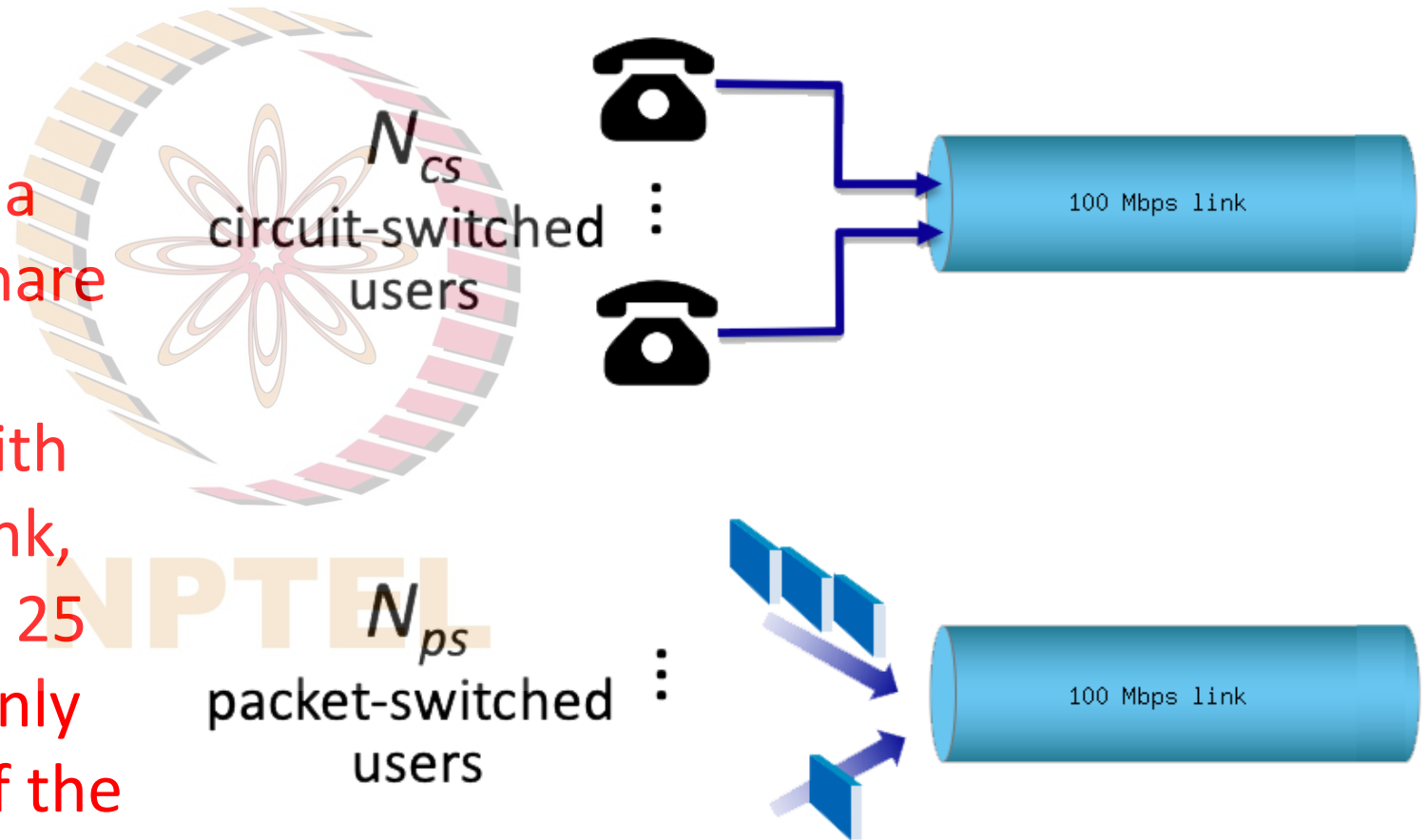


# Packet Switching



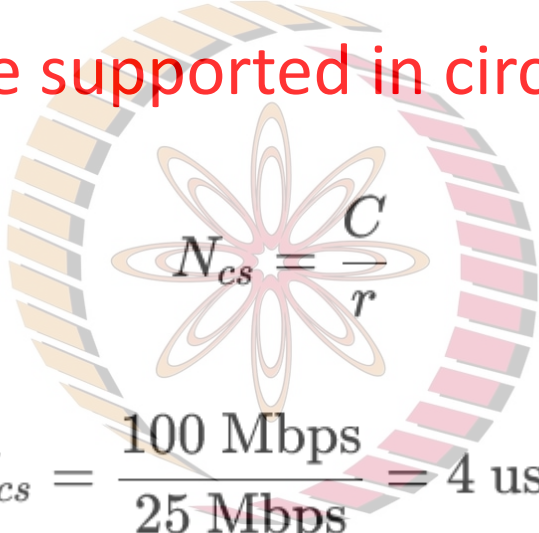
# Circuit Switching v/s Packet Switching

- A circuit-switching scenario in which  $N_{cs}$  users, each requiring a bandwidth of 25 Mbps, must share a link of capacity 100 Mbps.
- A packet-switching scenario with  $N_{ps}$  users sharing a 100 Mbps link, where each user again requires 25 Mbps when transmitting, but only needs to transmit 10 percent of the time.



# Circuit Switching v/s Packet Switching

I. How many users can be supported in circuit switching mode ?



The NPTEL logo is a circular emblem with a stylized flower or star shape in the center. The logo is composed of orange and pink segments. The text 'NPTEL' is written in large, bold, orange letters across the middle of the logo.

$$N_{cs} = \frac{C}{r}$$
$$N_{cs} = \frac{100 \text{ Mbps}}{25 \text{ Mbps}} = 4 \text{ users}$$

I. If there are 30 circuit switching users, can they be supported ?

No

# Circuit Switching v/s Packet Switching

❑ What is the probability that a given (specific) user is transmitting, and the remaining users are not transmitting?

The probability ( $P$ ) that a given specific user is transmitting and the remaining users are not is:

$$P = p \times (1 - p)^{N-1}$$

Substituting the values for  $N = 30$ :

$$P = 0.1 \times (0.9)^{30-1}$$

$$P = 0.1 \times (0.9)^{29}$$